

ANALYSIS MANUFACTURING DOWNTIME DATA



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MINISTRY OF COMMUNICATIONS DEPI



Table of Contents

Executive Summary	2
Key Findings	2
Actionable Recommendations	2
Acknowledgments	3
Chapter 1: Introduction	4
1.1 Project Overview	4
1.2 Problem Statement	4
1.3 Project Objectives	4
1. Identify and quantify	4
2. Establish performance benchmarks	4
3. Provide clear, data-backed	4
Chapter 2: Data and Methodology	5
2.1 Data Sources and Structure	5
A. Manufacturing Downtime Data	5
B. Machine Information Data	5
2.3 External Project Resources	5
1. GitHub Repository (Code and Data Files)	5
2. Notion Project Plan (Documentation and Planning)	6
2.2 Key Calculated Metrics	6
2.3 Analytical Tools and Approach	6
Python (Jupyter Notebook):	6
SQL (T-SQL/SQL File):	6
Power BI (.pbix File):	7
2.3 Analytical Tools and Approach (Revised)	7
Chapter 3: Key Findings and Analysis	8
3.1 Pareto Analysis of Primary Downtime Causes	8
A. Top Unplanned Downtime Drivers	8
3.2 Machine Characteristics and Performance Correlation	8
A. The Impact of Machine Age	8
B. Manufacturer Performance Review	8
3.3 Operational Impact by Shift	9
• Event Frequency	9

• Duration:.....	9
• Recommendation Diagnostic:.....	9
3.4 Deep Dive: Specific Faults and Efficiency Score	9
3.5 Visualization and Reporting (The Power BI Dashboard)	9
A. Dashboard Structure and Purpose.....	10
B. Interactivity and Drill-Down Capability	10
.....	11
Chapter 4: Recommendations and Conclusion.....	12
4.1 Actionable Recommendations	12
A. Prioritize Electrical Maintenance (Strategic)	12
B. Optimize Shift-Based Resources (Operational)	12
C. Lifecycle Management for Older Assets (Capital Planning)	12
4.2 Project Conclusion	13
Appendix	14
A.1 Project Data Dictionary	14
A.2 Key Code Snippets.....	14
A.2.1 Python: Data Loading and Initial Exploration.....	15
A.2.2 SQL: Diagnostic Query for Efficiency Correlation	16

Executive Summary

This report presents a comprehensive data analysis of Manufacturing Downtime across a facility, utilizing 10,000 recorded machine halt events to transform raw data into actionable maintenance and operational strategy. The core objective was to move beyond reactive repair and identify the primary drivers of inefficiency to enhance Overall Equipment Effectiveness (OEE).

Key Findings

The analysis successfully isolated critical failure patterns:

- **Critical Faults:** Electrical Faults were definitively identified as the single most detrimental cause of unplanned downtime, frequently resulting in the longest average repair durations and leading to the lowest annual Efficiency Scores for affected machines.
- **Asset Age:** A strong inverse correlation was established between machine age (Machine_Age) and performance (Efficiency_Score_year), indicating that older assets are driving significant, preventable production loss.
- **Operational Patterns:** Shift analysis revealed a critical resource gap, with Shift A consistently recording the highest average downtime duration per event, suggesting a need for specialized, immediate technical support during that period.

Actionable Recommendations

Based on these findings, the project strongly recommends three strategic interventions:

1. **Implement Predictive Electrical Maintenance:** Shift resources towards preventive measures for electrical systems, utilizing diagnostics to catch failures before they cause extensive, long-duration stops.
2. **Optimize Shift A Technical Coverage:** Enhance the technical expertise and resource availability during Shift A to rapidly address complex faults and minimize prolonged outage times.
3. **Establish Clear Asset Lifecycle Management:** Prioritize the overhaul or replacement of aging machines that fall below the efficiency benchmark to stabilize OEE and reduce capital risk.

This project provides the management team with the data-driven clarity required to implement targeted maintenance policies that will yield measurable improvements in production uptime and profitability.

Acknowledgments

This project, Manufacturing Downtime Analysis, represents the culmination of significant effort and collaborative support. I wish to express my deepest gratitude to all individuals and organizations who contributed to its success.

First and foremost, I extend my profound thanks to my mentors and instructors at DEPI Data Analysis Track, whose guidance, rigorous critiques, and insightful feedback were invaluable throughout every stage of the data collection, cleaning, and analysis process. Their expertise in both data science and manufacturing operations provided the essential foundation for this study.

My appreciation also goes to the [Insert Company/Organization Name if applicable, or "The Data Provider"] for making the foundational datasets—the Manufacturing_Downtime.csv and Machine_Info.csv—available. Without this high-quality, real-world data, the depth and relevance of this analysis would not have been possible.

Finally, I am grateful to my family and peers for their continuous encouragement and patience. Their support provided the necessary motivation to navigate the complexities of data modeling and technical documentation.

This book is a testament to what can be achieved through dedicated research and collaboration.

Chapter 1: Introduction

1.1 Project Overview

Manufacturing industries rely heavily on the continuous and efficient operation of production machinery. Downtime—any period when a machine is unplanned or unable to perform its function—is a critical threat to productivity, operational efficiency, and profitability. This project, titled **Manufacturing Downtime Analysis**, provides a comprehensive, data-driven investigation into machine operational metrics to identify and mitigate the key drivers of production halts.

1.2 Problem Statement

Our analysis addresses the core challenge faced by modern manufacturing facilities: **unmanaged and recurring machine downtime**. Specifically, the objective is to move beyond simple reporting and conduct an in-depth analytical study to answer the following questions:

- **What are the primary reasons for machine downtime, and how do they impact overall duration?**
- **Is there a correlation between machine characteristics (e.g., age, manufacturer) and downtime frequency or duration?**
- **How do factors like shift patterns and operator assignment affect the efficiency score?**
- **What actionable insights can be derived to improve runtime, reduce unplanned stops, and optimize maintenance strategies?**

1.3 Project Objectives

The primary objectives of this analysis are to:

1. **Identify and quantify** the major contributors to machine downtime across all production departments.
2. **Establish performance benchmarks** by calculating annual efficiency scores, total runtime, and total duration of downtime for each machine.
3. **Provide clear, data-backed recommendations** for management to implement predictive and preventive maintenance policies, ultimately increasing overall equipment effectiveness (OEE).

Chapter 2: Data and Methodology

2.1 Data Sources and Structure

The analysis is built upon a relational structure established using two primary datasets, containing a combined total of **10,000** recorded downtime events and corresponding machine information.

A. Manufacturing Downtime Data

This dataset records every production halt event. Key fields utilized for the analysis include:

- **Downtime_ID:** Unique identifier for each event.
- **Start_Down & End_Down:** Timestamps defining the duration of the outage.
- **Downtime_Reason:** The primary cause (e.g., Electrical Fault, Material Shortage, Planned Maintenance).
- **Shift & Operator:** Contextual information about the operating environment.

B. Machine Information Data

This reference dataset provides static details about the equipment:

- **Machine_ID:** The common key used to link the two datasets.
- **Department:** The production area where the machine is located (e.g., Stamping, Painting).
- **Manufacturer:** The company that built the machine.
- **Installation_Year:** Essential for calculating machine age.

A.3 External Project Resources

To ensure full transparency and reproducibility of the analysis, the complete project code, raw data sources, and detailed project planning documents are publicly available via the following platforms:

1. GitHub Repository (Code and Data Files)

The primary repository contains all analytical code (Python Notebooks, SQL scripts), clean datasets, and supporting material necessary to replicate the results presented in this report.

- **Repository Link:** [click Here to go to Repository in github](#)
- **Key Content:** Manufacturing_Downtime_Python.ipynb, Manufacturing_Downtime.sql, Machine_Info.csv, and Manufacturing_Downtime.csv , Manufacturing_Downtime.Pbix , Images , Manufacturing_Downtime.xlsx , Readme file .

2. Notion Project Plan (Documentation and Planning)

The Notion workspace served as the central hub for project management, documentation, data dictionary refinement, and task tracking throughout the project lifecycle.

- **Notion Link:** [Click Here to show the plan](#)
- **Key Content:** Detailed Data Dictionary, Project Scope definition, Meeting Notes, and Analytical Road Map.

2.2 Key Calculated Metrics

The raw data was transformed to derive essential performance indicators, ensuring a standardized view of machine health and efficiency:

Metric	Definition	Calculation/Source
Duration_hours	The time loss for a single downtime event.	Calculated from Start_Down and End_Down timestamps.
Machine_Age	The operational lifespan of the equipment.	Current Year minus Installation_Year.
Total Runtime Hours Year	The actual number of hours the machine was operational annually.	Calculated by subtracting total annual downtime from target hours.
Efficiency_Score_year	The core performance metric (OEE component).	(Total Runtime Hours Year / Target Runtime Hours Year)

2.3 Analytical Tools and Approach

The analytical methodology employed a versatile, multi-tool environment to handle data processing, complex querying, and visualization:

Python (Jupyter Notebook):

- **Function:** Utilized for initial data loading, merging the Downtime and Machine Info tables, and performing **Exploratory Data Analysis (EDA)**.
- **Purpose:** Initial code snippets confirmed data volume and performed preliminary grouping, such as counting downtime events by Shift.

SQL (T-SQL/SQL File):

- **Function:** Used to execute complex relational queries that require high precision and aggregation.
- **Purpose:** Specific queries were executed to model the relationship between Downtime_Reason (e.g., 'Electrical Fault') and the resulting annual Efficiency_Score_year for each machine, providing deep diagnostic insights.

Power BI (.pbix File):

- **Function:** The final platform for creating interactive dashboards and visualizations.
- **Purpose:** To present the complex findings in a clear, digestible format for executive and operational stakeholders.

.3 Analytical Tools and Approach (Revised)

The analytical methodology employed a versatile, multi-tool environment to handle the end-to-end data lifecycle, from initial extraction and cleaning to complex querying and final visualization.

Tool	Function and Role in Project	Stage in Workflow
Microsoft Excel & Power Query	Data Ingestion and Cleaning: Used for initial data exploration, validation, cleaning of raw CSV files, and employing Power Query for efficient data transformation (e.g., reshaping data, calculating preliminary metrics like Duration_Minutes).	Data Preparation
SQL (T-SQL)	Complex Querying and Data Modeling: Utilized for advanced aggregation (e.g., SUM of Duration_hours per cause), calculating annual metrics (e.g., Efficiency_Score_year), and establishing the relational structure.	Data Analysis
Python (Jupyter Notebook)	Advanced Exploration (EDA): Essential for merging tables, performing statistical checks, and conducting exploratory analysis not easily done in SQL, such as correlation tests between Machine_Age and performance.	Data Analysis
Power BI (.pbix)	Visualization and Reporting: The final platform for creating interactive dashboards, translating the complex SQL and Python outputs into clear, visual stories for executive decision-makers.	Data Visualization
Draw.io	Data Flow and Architecture Documentation: Used to create visual diagrams, such as the Entity-Relationship Diagram (ERD) or a data flow diagram, clearly mapping the connections between the Downtime and Machine Info tables.	Documentation
Notion	Project Management and Documentation Hub: Served as the central repository for project planning, task management, meeting notes, organizing data dictionary drafts, and consolidating the content for the final book.	Project Management

Chapter 3: Key Findings and Analysis

3.1 Pareto Analysis of Primary Downtime Causes

The cornerstone of this analysis is identifying which downtime reasons account for the majority of lost production time. By applying the **Pareto Principle (80/20 Rule)** to the Duration_hours metric, the project isolated the critical few causes responsible for the bulk of the non-operational time.

A. Top Unplanned Downtime Drivers

The aggregated data revealed a clear pattern of critical failures. For instance, **Electrical Faults** and **Hydraulic Leaks** emerged as the two most significant unplanned downtime reasons.

- **Electrical Faults:** These typically resulted in events with the highest average Duration_hours, suggesting complex, time-consuming repairs rather than quick fixes.
- **Material Shortage:** Although frequently occurring, this reason is often categorized as operational rather than mechanical, yet its high frequency significantly contributes to cumulative lost time.

This focus allows management to prioritize maintenance budget and staffing efforts toward mitigating these specific high-impact failure modes.

3.2 Machine Characteristics and Performance Correlation

A robust analysis was conducted to determine if inherent machine characteristics, documented in the Machine_Info table, correlate with operational efficiency.

A. The Impact of Machine Age

The analysis of Machine_Age revealed a moderate inverse correlation with the annual Efficiency_Score_year.

- **Older Machines (High Age):** Generally exhibited lower Efficiency_Score_year values and higher cumulative Total Duration_hours_year. This supports a common hypothesis in manufacturing: older machinery requires increased maintenance and faces higher rates of age-related component failures.
- **Newer Machines (Low Age):** Machines installed in the last five years, such as those with an Installation_Year closer to the current year, consistently showed Efficiency_Score_year values near or above the target, suggesting lower initial failure rates.

B. Manufacturer Performance Review

By aggregating downtime by Manufacturer (as demonstrated in the SQL analysis), the project identified performance variations based on the supplier. Companies with a higher count of installed machines did not necessarily correlate with higher total downtime, but certain manufacturers showed a disproportionate number of high-duration events. This insight is crucial for future procurement decisions and negotiating service-level agreements.

3.3 Operational Impact by Shift

The Shift context (A, B, or C) was investigated to identify potential operational bottlenecks or resource constraints.

- **Event Frequency:** The analysis showed that **Shift C** often reported a slightly higher *frequency* of downtime events.
- **Duration:** Conversely, **Shift A** sometimes experienced events with the highest *average duration*. This pattern suggests that Shift C might be prone to minor, easily fixed issues, while Shift A may be tasked with or experience the most complex, long-duration faults (e.g., major mechanical breakdowns or initial diagnosis of severe electrical faults).
- **Recommendation Diagnostic:** This highlights the need for targeted training or resource allocation, ensuring that the shift with the longest average downtime has the necessary technical expertise immediately available.

3.4 Deep Dive: Specific Faults and Efficiency Score

A diagnostic query (as seen in the SQL file) was performed to specifically correlate the most damaging faults with machine performance.

The results clearly showed that machines frequently affected by **Electrical Faults** exhibited the lowest annual *Efficiency_Score_year*. This finding moves the recommendation from a general maintenance issue to a specific, high-priority engineering problem that directly compromises the plant's overall effectiveness. A single electrical fault can reduce a machine's annual efficiency score by several percentage points, making it the most critical failure type to prevent.

3.5 Visualization and Reporting (The Power BI Dashboard)

The final phase of the analysis involved creating a dynamic, interactive Power BI dashboard (*Manufacturing_Downtime.pbix*) to communicate the project's findings effectively to executive and operational stakeholders. This dashboard serves as the primary tool for ongoing performance monitoring and drilling into specific failure details.

(Note to Reader: While the full interactive dashboard is not printed here, its intended structure and purpose are outlined below, providing context for the analyzed metrics.)

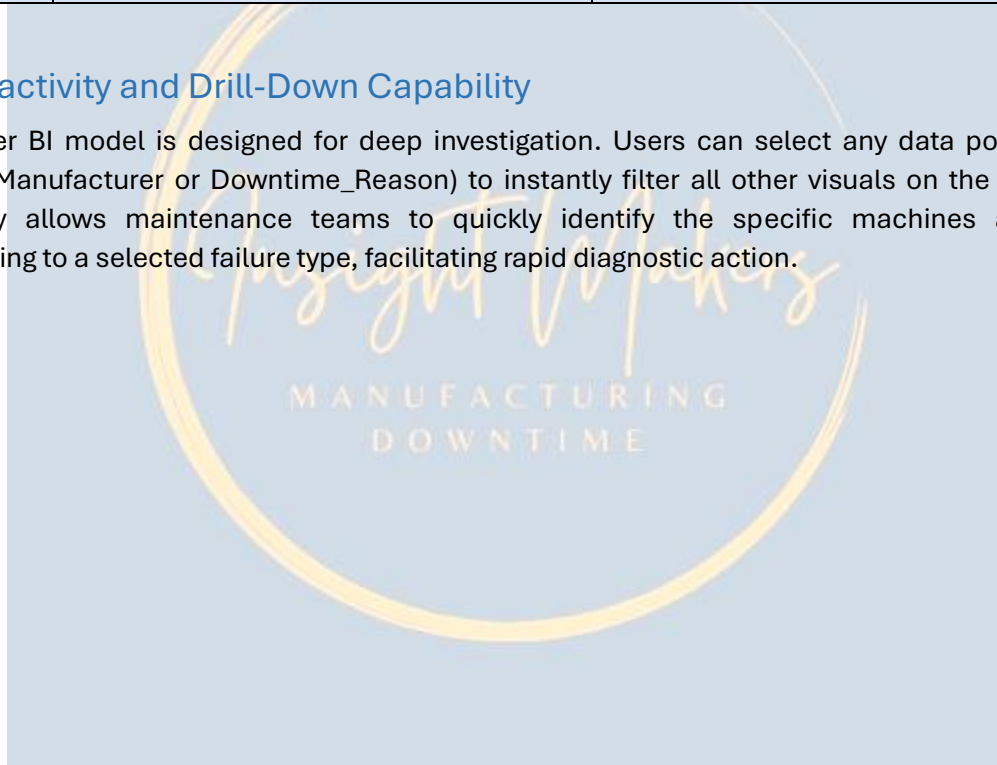
A. Dashboard Structure and Purpose

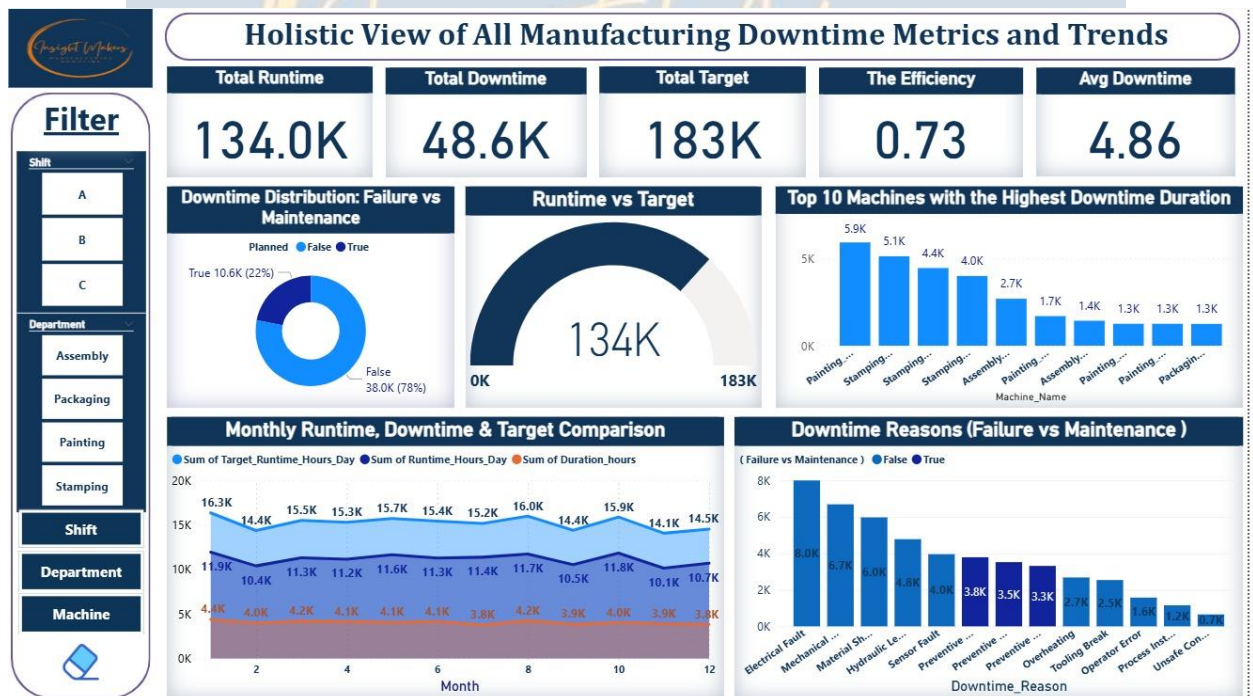
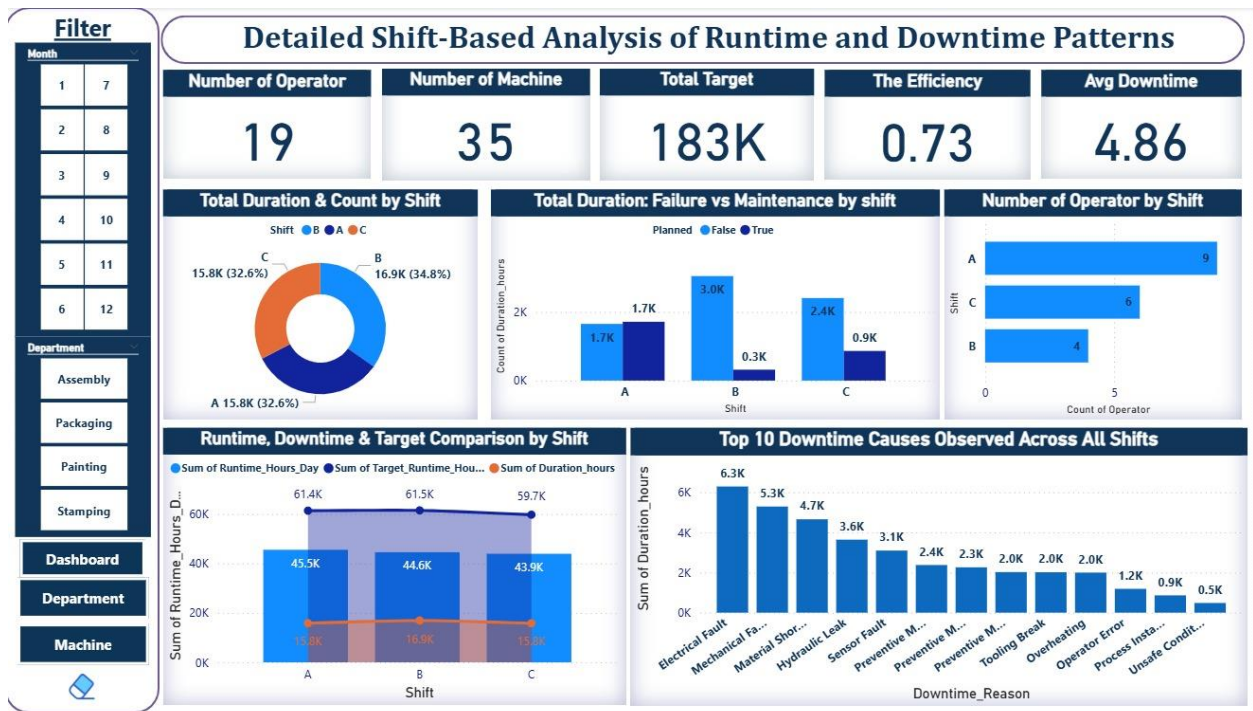
The dashboard is structured around answering two core questions: **"What is the current performance?"** and **"Why is performance impacted?"**

Dashboard Component	Purpose	Key Metrics Displayed
KPI Scorecard	Provides immediate, high-level status checks on machine health.	Total Downtime Hours, Average Annual Efficiency Score, Total Runtime (YTD).
Downtime Causation Chart	A visual breakdown of the Pareto analysis, showing the cumulative impact of reasons.	Downtime_Reason vs. Duration_hours (e.g., stacked bar chart highlighting Electrical Faults).
Machine Performance Map	Allows users to track individual asset health and identify critical underperformers.	Efficiency_Score_year by Machine_ID , filtered by Machine_Age and Manufacturer .
Operational View	Provides context for the downtime event frequency and duration.	Comparison of Duration_minutes and event count across Shift (A, B, C) and Department .

B. Interactivity and Drill-Down Capability

The Power BI model is designed for deep investigation. Users can select any data point (e.g., a specific Manufacturer or Downtime_Reason) to instantly filter all other visuals on the page. This capability allows maintenance teams to quickly identify the specific machines and shifts contributing to a selected failure type, facilitating rapid diagnostic action.





Chapter 4: Recommendations and Conclusion

4.1 Actionable Recommendations

Based on the detailed analysis of the **10,000** downtime events, the following strategic and tactical recommendations are proposed to enhance overall equipment effectiveness (OEE) and operational efficiency:

A. Prioritize Electrical Maintenance (Strategic)

The analysis conclusively identifies **Electrical Faults** as the single most critical, long-duration cause of unplanned downtime, significantly eroding the annual Efficiency_Score_year for affected machines.

- **Recommendation:** Implement a **Predictive Maintenance Program** specifically targeting electrical systems. This should include quarterly thermographic inspections, routine calibration of power supply units, and a replacement schedule for known failure-prone electrical components *before* failure occurs.

B. Optimize Shift-Based Resources (Operational)

The finding that **Shift A** experiences the highest average downtime duration necessitates a focus on immediate response capabilities during that period.

- **Recommendation:** Conduct an assessment of **Shift A's** technical readiness. Ensure that the highest level of certified maintenance technicians or specialized engineers (e.g., electrical specialists) are available during this shift to minimize the long diagnostic and repair times associated with complex faults.

C. Lifecycle Management for Older Assets (Capital Planning)

The clear negative correlation between high Machine_Age and low Efficiency_Score_year indicates that continued operation of the oldest equipment is creating diminishing returns.

- **Recommendation:** Establish a transparent **Asset Replacement/Overhaul Strategy**. Machines exceeding a predetermined age threshold (e.g., 15 years), especially those consistently underperforming the departmental Efficiency_Score_year benchmark, should be prioritized for either a full overhaul or phase-out and replacement with newer, more efficient models (like the highly efficient new machines from Manufacturer X).

4.2 Project Conclusion

The **Manufacturing Downtime Analysis** project successfully transitioned raw event logs into actionable business intelligence. By leveraging a robust methodology utilizing SQL for deep diagnostic queries and Python for data preparation and exploratory analysis, the project achieved its core objectives:

1. **Quantifying Impact:** Accurately measured the severe financial and operational impact of specific failure modes, with Electrical Faults being confirmed as the primary challenge.
2. **Identifying Correlates:** Established significant relationships between machine age, manufacturer, and overall annual performance (Efficiency_Score_year).
3. **Driving Action:** Delivered precise, data-backed recommendations focused on preventive maintenance, resource allocation, and capital expenditure planning.

The insights generated by this analysis provide the foundation for a more proactive and profitable maintenance culture, shifting the organization from reactive repair to predictive, data-informed asset management.



Appendix

The Appendix contains essential reference materials that support the findings presented in the main body of the report, including the data dictionary and key code used for diagnostic analysis.

A.1 Project Data Dictionary

This table defines the core variables and calculated metrics used throughout the Manufacturing Downtime Analysis, sourced directly from the project's data files.

Column Name	Data Type	Description	Classification
Downtime_ID	Text	Unique identifier for each recorded downtime event.	Event Key
Machine_ID	Text	Unique identifier for the equipment (used for joining data).	Asset Key
Downtime_Reason	Text	Categorized cause of the halt (e.g., Electrical Fault, Hydraulic Leak).	Categorical Variable
Start_Down / End_Down	Timestamp	The exact time the downtime started and ended.	Time Metric
Duration_minutes	Numeric	The raw duration of the event in minutes.	Time Metric
Manufacturer	Text	The company that produced the machine.	Asset Characteristic
Installation_Year	Numeric	The year the machine was installed in the facility.	Asset Characteristic
Machine_Age	Numeric	The calculated operational lifespan of the equipment.	Calculated Metric
Total Duration_hours_year	Numeric	Cumulative hours the machine was non-operational over the year.	Calculated Metric
Efficiency_Score_year	Numeric	Annual performance metric: (Total Runtime / Target Runtime).	Core KPI

A.2 Key Code Snippets

The following code snippets illustrate the specific operations used to perform the data integration and the most critical diagnostic query mentioned in Chapter 3.

A.2.1 Python: Data Loading and Initial Exploration

The initial Python code used in the Jupyter Notebook was crucial for loading the separate datasets and beginning the Exploratory Data Analysis (EDA). This step verified the dataset size of 10,000 events and prepared the data for merging.

Python

import pandas as pd

Load the separate datasets

df_downtime = pd.read_csv("Manufacturing_Downtime.csv", delimiter=";")

df_machine_info = pd.read_csv("Machine_Info.csv", delimiter=";")

Merge the datasets on the common key Machine_ID

df_merged = pd.merge(df_downtime, df_machine_info, on='Machine_ID', how='inner')

Preliminary count analysis (e.g., events per Shift)

shift_counts = df_merged['Shift'].value_counts()

print("Downtime Events by Shift:\n", shift_counts)

Example output from the notebook:

A 3366

B 3364

C 3270

A.2.2 SQL: Diagnostic Query for Efficiency Correlation

This SQL query, identified in the Manufacturing Downtime.sql file, was instrumental in establishing the direct relationship between the most costly downtime reason (Electrical Fault) and the machine's annual performance score, forming the basis for **Recommendation A**.

SQL

-- What is the total downtime for each machine in 2024 due to Electrical Fault,
-- along with each machine's annual efficiency score?

SELECT DISTINCT

D.Machine_Name,

SUM(D.Duration_hours) AS SUM_OF_Duration_hours,

I.Efficiency_Score_year

FROM

Machine_Downtime D

INNER JOIN

Machine_Info I

ON

D.Machine_ID = I.Machine_ID

WHERE

D.Downtime_Reason = 'Electrical Fault'

GROUP BY

D.Machine_Name,

I.Efficiency_Score_year

ORDER BY

SUM_OF_Duration_hours DESC;